

Brieuc Collet, Researcher in Space Physics

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🔍 ADS Page

Space physics researcher specialized in planetary wave-plasma instability

Employment History

- 2025– 2027  **Post Doc** PPARC, Tohoku University, Japan
Understanding Mercury and the Jovian environment with in situ radio measurements from Juno, BepiColombo and Juice
- 2022– 2025  **PhD in Astrophysics** Lab of Astrophysics of Marseille, Aix-Marseille University, France
Understanding the microphysics of Jovian radio auroral emissions with Juno space probe

Education

- 2022 – 2025  **PhD in Astrophysics, Aix-Marseille Université**
- 2021 – 2022  **Master's Degree Plasma Physics, Paris Saclay University**
- 2019–2020  **Bachelor's Degree of Fundamental Physics, Paris Saclay University**

Teachings

- 2022-2025  **Teaching Assistant**, 192h, Aix-Marseille University

Scientific Outreach

- 2022–2025  **Festival of Astronomy of Provence**
-  **Fête de la Science - French Science Diffusion Day**

Skills

- Programming  Proficiency in **Python**, Knowledge in **C++** and **IDL**
- Astronomy  Observation nights at Observatory of Haute Provence

Community Activities

- 2022–2025  **Laboratory's Sustainable Development Group Member**
-  **Laboratory's Seminar Organizing Committee**
- 2023–2025  **PhD Representative in Laboratory Council**
- 2025  **Local Organizing Committee** for PREX, Marseille France
- 2025-2026  **Reviewer** for PREX proceedings and Geophysical Research Letters
-  **Associate Scientist** BepiColombo/Mio/PWI/Sorbet

Awards

- 2022  PhD grant
- 2025  **Best PhD presentation award**, Planetary, solar and heliospheric Radio Emissions X, Marseille, France

Publications

Articles

- [1] Al Saati, S., Clément, N., [...], **Collet, B.**, et al. 2022. "Magnetosphere -Ionosphere -Thermosphere Coupling Study at Jupiter Based on Juno's First 30 Orbits and Modeling Tools". In: *Journal of Geophysical Research: Space Physics*. [DOI: 10.1029/2022JA030586](#).
- [2] Benmahi, B., Hue, V., [...], **Collet, B.**, et al. 2025. "Precipitating electron properties of Ganymede's aurora retrieved from Juno/UVS observations during P34". In: *Astronomy & Astrophysics*. [DOI: 10.1051/0004-6361/202556996](#).
- [3] **Collet, B.**, Lamy, L., et al. 2023. "Characterization of Jovian hectometric sources with Juno: statistical position and generation by shell-type electrons". In: *PRE IX*. Ed. by C. K. Louis, C. M. Jackman, et al. DIAS and TCD. [DOI: 10.25546/103095](#).
- [4] **Collet, B.**, Lamy, L., et al. 2024. "A New Type of Jovian Hectometric Radiation Powered by Monoenergetic Electron Beams". In: *Journal of Geophysical Research: Space Physics*. [DOI: 10.1029/2024JA032422](#).
- [5] **Collet, B.**, Lamy, L., et al. 2025. "In Situ Analysis of Jupiter's Broadband Kilometric Auroral Radio Emissions With Juno". In: *Geophysical Research Letters* 52.9. [DOI: 10.1029/2024gl114447](#).
- [6] **Collet, B.**, Lamy, L., et al. 2026. "Detection of Jovian kilometric and hectometric auroral radio harmonics with Juno/Waves in situ measurements". In: *Planetary, Solar and Heliospheric Radio Emissions X*. Ed. by L. Lamy, C. K. Louis, et al. Accepted, preprint. OSU Pythéas/AMU, Observatoire de Paris. [DOI: 10.25935/PREX-TR41](#).
- [7] Heidari, N., Boisse, I., [...], **Collet, B.**, et al. 2024. "The SOPHIE search for northern extrasolar planets: XIX. A system including a cold sub-Neptune potentially transiting a $V = 6.5$ star HD 88986". In: *Astronomy & Astrophysics*. [DOI: 10.1051/0004-6361/202347897](#).
- [8] Lamy, L., Cecconi, B., [...], **Collet, B.**, et al. 2023. "Re-exploring the radio spectrum of Uranus in orbit: science case and digital high-frequency receiver". In: *PRE IX*. Ed. by C. K. Louis, C. M. Jackman, et al. DIAS and TCD. [DOI: 10.25546/103105](#).
- [9] Louis, C. K., Boudouma, A., [...], **Collet, B.**, et al. 2026. "Search for the influence of the Galilean moons on Jovian kilometric and decametric radio emissions over 7 years of Juno data". In: *Planetary, Solar and Heliospheric Radio Emissions X*. Ed. by L. Lamy, C. K. Louis, et al. OSU Pythéas/AMU, Observatoire de Paris. [DOI: 10.25935/PREX-VWFQ](#).
- [10] Louis, C. K., Louarn, P., **Collet, B.**, et al. 2023. "Source of Radio Emissions Induced by the Galilean Moons Io, Europa and Ganymede: In Situ Measurements by Juno". In: *Journal of Geophysical Research: Space Physics*. [DOI: 10.1029/2023JA031985](#).
- [11] Rabia, J., Hue, V., [...], **Collet, B.**, et al. 2025. "In situ and remote observations of the ultraviolet footprint of the moon Callisto by the Juno spacecraft". In: *Nature Communications*. [DOI: 10.1038/s41467-025-62520-4](#).
- [12] Wu, Siyuan, Whiter, Daniel K., [...], **Collet, Brieuc**, et al. 2026. "A radio oval above Earth's auroral oval". In: *Science Advances*. [DOI: 10.1126/sciadv.aec4114](#).
- [13] **Collet, B.** 2025. "Understanding the microphysics of Jovian radio auroral emissions with Juno". PhD thesis supervised by Lamy, L. at Laboratoire of Astrophysics of Marseille, Aix-Marseille University. PhD thesis.

Conferences

- 2022  **French National Sun-Planet meeting (ATST)**, Marseille, France
-  **Planetary, solar and heliospheric Radio Emissions IX**, Dublin, Ireland
- 2023  **Chapman Understanding Alfvén Waves in the Sun and the Heliosphere**, Berlin, Germany
- 2024  **Astroradiofr24, French radio frequency astrophysics, towards SKA**, Paris
-  **French National Sun-Planet meeting (ATST)**, Marseille, France
-  **Magnetospheres of Outer Planets**, Minneapolis, USA
- 2025  **EGU General Assembly**, Vienne, Austria
-  **Planetary, solar and heliospheric Radio Emissions X**, Marseille, France
- 2026  **Symposium on Planetary Sciences**, Sendai, Japan
-  **JPGU**, Chiba, Japan (next May)
-  **French National Sun-Planet meeting (ATST)**, Biarritz, France
-  **Magnetospheres of Outer Planets**, Toulouse, France (next July)